

Control Flow

The adventure really starts here. This leg of the journey may be challenging, but you need to make sure you really understand these concepts and how to use them. Control flow is the central mechanism in programs when viewed as instructions. A solid understanding of the concepts and how to apply them will position you to speed through the next few modules.

Learn to use control flow concepts and tools to create programs that respond to inputs and events. This will greatly enhance what you are capable of achieving. You should see this as a major milestone.

Key aspects include:

- Seeing how boolean logic is used to control.
- Remembering that everything is still just a sequence, and that loops and branches simply adjust which action is performed next.
- Learn to apply sequence, selection (branching), and repetition (loops) to build small programs

Show us that you can:

- Create small programs with:
 - Boolean expressions that include comparisons and logical operators to form expressions that test given conditions
 - Branches using *if* and *switch/case* statements
 - Loops using *while*, *do while*, and *for* statements
- Correctly use programming terminology to explain your code.
- Do this task **without** using any concepts we haven't covered yet.

- Capture your learning in the Learning Journey section. Include evidence (screenshots) of your progress, and your thoughts on what you accomplished, what challenges you faced, and what you found interesting.
- Provide sincere *reflections* in the Reflection section. Don't just tell us what you did, tell us what you *learned*. Reflections that are brief or shallow will not be accepted.

Tip

This is a big and important topic! You will need to dedicate some time to this, and you need to make sure that you really understand these ideas as they are central to everything in programming. Ensure you make use of all the support available to help you learn to use these new tools.

Complete the following:

1. Download and open the template from OnTrack.
2. Work your way through the **Overview**, **Guided Tour**, and **In Depth** sections from [Chapter 3: Control Flow](#) of the V1 Programmers Guide.
3. Demonstrate how the following two small programming snippets run using the [Hand Execution](#) process as described in the In Depth section. Do this on paper with a pencil. **Capture** a photo of your workings.

```
using static System.Console;
using static System.Convert;

int target;

Write("Count to: ");
target = ToInt32(ReadLine()); // User enters 5

int i = 0;

while (i < target)
{
    WriteLine(i + 1);
    i++;
}
```

```
using static System.Console;
using static System.Convert;

int i;

for (i = 0; i < 8; i++)
{
    if (i > 0 && i % 3 == 0)
    {
        i++;
        continue;
    }

    WriteLine(i);
}
```

4. Complete one of the [Test Your Knowledge](#) activities from the Control Flow chapter of the Programmers Field Guide.
5. You should have enough to submit now, but if you want to deepen your understanding have a look at [Chapter 2: Communicating Syntax](#).

Tip: A bit stuck? Be sure to reach out!

Your tutors are programming experts and are there to help. Be sure to reach out to them during the seminar, at the HelpHub, or ask a question on the SIT102 Teams channel.

6. Complete your submission – complete the summary and related activities in the template.
7. **Upload** to OnTrack and **engage with feedback** to get this complete!

Available Resources

We have many resources to assist in your learning journey – if you are ever unsure then please reach out to us, we are always happy to help.

- **The SIT102 Teams channel** – an active learning resource with weekly announcements from the Unit Chair, and regular discussions/assistance from the SIT102 teaching team and students.
- **Seminars** – you can attend both on-campus and online sessions, and join as many as you like each week. Your tutors will provide demonstrations and assistance to help guide your learning.
- **HelpHubs** – there are sessions hosted nearly everyday, both on-campus and online providing one-on-one assistance from a tutor. Please join as many as you like.
- **PASS sessions** – running every week, these sessions are run like a study group, hosted by a past SIT102 student.

For more information about the above resources please speak to your tutor or visit the SIT102 Teams channel.